VitalTwin — AI-Powered Clinical Digital Twin Architecture

End-to-End Longitudinal Risk Prediction, Counterfactual Simulation & Governance Pipeline

1. Clinical Ingestion Layer

- Synthea Synthetic Cohort (1,335 pts)
- 6,451 Longitudinal Encounters
- ABDM FHIR R4 Bundle Parser
- LOINC: 8480-6 (SBP), 8462-4 (DBP)
- Simulated Bluetooth Vitals & SpO2
- Server-Sent Events (SSE) Live Feed
- Strict Zero-PHI / Synthetic Guardrails

2. Feature Engineering & State

- 132 Engineered Longitudinal Features
- Strict 365-Day Rolling Lookback
- Zero Target-Leakage Temporal Split
- Pulse Pressure (SBP - DBP)
- Trajectory Slopes & BP Volatility
- ColumnTransformer Pipeline
- Schema Fingerprint: a4ddaedbdbc8bb07

3. AI/ML & Explainability Core

- XGBoost Classifier Ensemble
- 365-Day Decompensation Target
- Platt Logistic Calibrator (95% CI)
- TreeSHAP Signed Feature Drivers
- ROC-AUC: 0.8069 | PR-AUC: 0.0644
- Correlated Simulator (lever_map.json)
- Alert-Budget Operating Policies

4. Security, Audit & Surveillance

- Append-Only SHA-256 Cryptographic Hash Chaining
- Inference & Simulation Block Verification Walk
- DPDP Act 2023 Principles: Consent & Minimization
- Early Warning Trajectory Surveillance & Lead Time Days
- Out-of-Distribution Bounds Checking & Clamping

5. Next.js 16 Clinical Workspace

- Dual 3D Spatial Viz: Abstract Lattice & Anatomical Avatar
- Organ Hotspots (Brain, Heart, Kidneys, Arteries)
- Multilingual Patient View (English, Hindi, Kannada)
- What-If Counterfactual Lab with Correlated Levers
- Cohort Directory, Trajectory Sparklines & PDF Export